Prob. 8. (Z & T, Ch. 7. 8.4.) piecewise smooth $\Rightarrow$ uniform convergence.

Let $f(x)$ be sectionally continuous on $[-\pi, \pi]$.

(a) If $f(x)$ is an even f-n show that its F.S. is even, (i) and (ii) that its Fourier coef-s are given by

(1) $$a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cdot\cos(nx)\,dx \quad , n = 0, 1, 2, \ldots;$$

(2) $$b_n = 0 \quad , \quad n = 1, 2, 3, \ldots$$

Sol-n:

(i) Let $f(x)$ be an even f-n. By def-n of even f-n, then

(3) $$f(-x) = f(x).$$

Also, recall the F.S of $f(x)$.

(4) $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\cdot\sin(nx)\right]$$ & consider

(5) $$f(-x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left\{a_n\cos(n(-x)) + b_n\sin(n(-x))\right\}.$$

Yet by (3) $\equiv$ (4), so can add the LHS & RHS's of both:

$$f(x)+f(-x) = \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right] + \frac{a_0}{2} + \frac{a_0}{2} + \sum_{n=1}^{\infty}\left\{a_n\cdot\cos(-nx) + b_n\cdot\sin(-nx)\right\} \Rightarrow$$

$$\underbrace{f(x)+f(-x)}_{=f(x),\text{ by (3)}} = 2\cdot\frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + \cancel{b_n\cdot\sin(nx)} + a_n\cos(-nx) + \cancel{b_n\sin(-nx)}\right] \Rightarrow$$

$$f(x)+f(x) = 2f(x) = a_0 + \sum_{n=1}^{\infty}\left[2a_n\cos(nx)\right]$$ as $\boxed{\begin{array}{l}\sin(-nx) = -\sin(nx)\\ \cos(-nx) = \cos(nx)\end{array}}$

(5) $\Rightarrow$ $$\underbrace{\boxed{f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n\cos(nx)}}_{F.S.(-x) = F.S.(x)}$$ establishing that F.S. of even f-n $f(x)$ is even.

(ii) Now in the next part, WNT:

(1) $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(nx)\,dx$, $n = 0, 1, 2 \ldots$

(2) $b_n = 0$, $n = 1, 2 \ldots$

Note $f(x) = f(-x)$ & the FS for $f(x)$ on $[-\pi, \pi]$ has been given in Pr(6):

(7) $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n\cos(nx) + b_n\sin(nx)\right]$ $\Big\}\cdot\cos(mx)$ of BS & $\int_{-\pi}^{\pi}$: [assuming $\int\sum = \sum\int$]

$$\int_{-\pi}^{\pi} f(x)\cos(mx)\,dx = \int_{-\pi}^{\pi}\cos(mx)\cdot\frac{a_0}{2}\,dx + \sum_{n=1}^{\infty}\left\{a_n\underbrace{\int_{-\pi}^{\pi}\cos(nx)\cdot\cos(mx)\,dx}_{=0 \text{ if } n\neq m} + b_n\underbrace{\int_{-\pi}^{\pi}\sin(nx)\cos(mx)\,dx}_{=0,\text{ always by Pr.(5)}}\right\}$$

$$\Rightarrow \int_{-\pi}^{\pi} f(x)\cdot\cos(mx)\,dx = \underbrace{\frac{a_0}{2}\int_{-\pi}^{\pi}\cos(mx)\,dx}_{=0,\text{ by Pr.5(a)}} + a_m\underbrace{\int_{-\pi}^{\pi}\cos(mx)\cdot\cos(mx)\,dx}_{=\pi,\text{ by the result of Pr.(5), (f)}} + 0 \Rightarrow$$

$$\Rightarrow \int_{-\pi}^{\pi} f(x)\cos(mx)\,dx = 0 + a_m\cdot\pi \Rightarrow$$

(8) $$\boxed{a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cdot\cos(mx)\,dx}$$ for $m = 0, 1, 2, 3, \ldots$ (should work by the result of Pr.7.)

But, note that, in (8), $f(x)$ = even f-n, & $\cos(mx)$ = even f-n, so must be their product, i.e. $f(x)\cdot\cos(mx)$ = even f-n. Then, by the symetric integral of even f-n:

$$\int_{-\pi}^{\pi}\text{even}(x)\,dx = \text{odd}(x)\Big|_{-\pi}^{\pi} = \text{odd}(\pi) - \text{odd}(-\pi) = \text{odd}(\pi) + \text{odd}(\pi) = 2\,\text{odd}(\pi)$$

$$= 2\cdot\int_0^{\pi}\text{even}(x)\,dx. \quad \text{i.e.}$$

(9) $\int_{-\pi}^{\pi}\text{even}(x)\,dx = 2\int_0^{\pi}\text{even}(x)\,dx$, applying to (8) gives the desired result:

(10) $a_m = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(mx)\,dx$ for $m = 0, 1, 2 \ldots$

For $b_n = 0$, by the result of Pr. (6).

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)}_{\text{even}}\,\underbrace{\sin(nx)}_{\text{odd}}\,dx = \frac{1}{\pi}\int_{-\pi}^{\pi} \text{odd}(x)\,dx = 0 \quad \left[\text{like } \int_{-\pi}^{\pi} 2x\,dx = 0\right]$$

(11) $\Rightarrow b_n = 0 \quad \forall n = 1, 2, 3 \ldots$ ▨

(b) If $f(x)$ is an odd f-n show that (i) its F.S. is odd & (ii) that its F.S. coef-s are

$$a_n = 0 \quad n = 0, 1, 2 \ldots, \qquad b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)\,dx, \quad n = 1, 2 \ldots$$

Sol-n: This case is only the reverse (odd replaced the even in part (a), & all claims are reversed by odd) version of (a), so let me be briefer:

(i) Let $f(x)$ be odd:

(12) $$f(-x) = -f(x) \Rightarrow \boxed{f(x) + f(-x) = 0}$$

Then by (4), F.S.:

(13) $$\begin{cases} f(x) = \frac{a_0}{2} + \sum\left[a_n\cos(nx) + b_n\sin(nx)\right] \\ f(-x) = \frac{a_0}{2} + \sum\left[a_n\cos(-nx) + b_n\sin(-nx)\right] \end{cases} \quad \&$$

$$0 = 2\cdot\frac{a_0}{2} + \sum 2a_n\cos(nx) + 0 \Rightarrow$$ (Considering an infinite dim-l vector space, basis ⊥)

$$\sum_{n=1}^{\infty} a_n\cos(nx) + \frac{a_0}{2} = \sum_{n=1}^{\infty} 0\cdot\cos(nx) + 0 \Rightarrow$$

$a_n = 0 \quad \forall n = 1, 2 \ldots$ & $\frac{a_0}{2} = 0 \Rightarrow a_0 = 0 \Rightarrow$

$a_n = 0 \quad \forall n = 1, 2, 3 \ldots$ (we could show this by $a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)}_{\text{odd}}\underbrace{\cos(nx)}_{\text{even}}\,dx = \int_{-\pi}^{\pi} \text{odd}(x)\,dx = 0$ like above.)

Conversely, for (13),

$$-\begin{cases} f(x) = \frac{a_0}{2} + \sum\left[a_n \cos(nx) + b_n \cdot \sin(nx)\right] \\ f(-x) = \frac{a_0}{2} + \sum\left[a_n \cos(-nx) + b_n \cdot \sin(-nx)\right] \end{cases}$$

$$f(x) - \underbrace{f(-x)}_{-f(x)} = \sum b_n \sin(nx) - b_n \sin(-nx) \quad \Rightarrow \div \frac{1}{2}$$

(14) $$\underbrace{\boxed{f(x) = \sum_{n=1}^{\infty} b_n \sin(nx)}}_{F.S(-x) = -F.S.(x) \text{ because } \Sigma \text{ of odd } (\sin(nx)) \text{ f-ns.}}$$

(ii) For the next part WNT:

(15) $a_n = 0, \quad n = 1, 2, 3 \ldots$ &

(16) $$b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin(nx)\,dx \qquad n = 1, 2 \ldots$$

By (6) in part (a), here $f(-x) = -f(x)$, so

(17) $$a_m = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)}_{=\text{odd}} \cdot \underbrace{\cos(mx)}_{\text{even}} \cdot dx = \int_{-\pi}^{\pi} \text{odd}(x)\,dx \equiv 0 \quad \text{for } m = 0, 1, 2 \ldots$$

And by the result of Pr. 6:

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{f(x)}_{\text{odd}} \underbrace{\sin(nx)}_{\text{odd}}\,dx = \int_{-\pi}^{\pi} \text{even}(x)\,dx = 2\int_0^{\pi} \text{even}(x)\,dx =$$

$$= 2 \cdot \frac{1}{\pi}\int_0^{\pi} f(x) \sin(nx)\,dx \Rightarrow \text{i.e.}$$

$$b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin(nx)\,dx \Rightarrow \text{proving all asked.} \quad \blacksquare$$